# Appendix: Supplementary Tables and Figures

Table A1 reports the 10-repeat split summary (mean ± std), and Table A2 reports a post-hoc split-conformal calibration check on the fixed seed-42 protocol. Figure A1 shows category balance together with the orientation-colored joint target view, Figure A2 shows residuals versus predictions, and Figure A3 shows standardized residuals with the reliability curve.

**Table A1.** Repeated split evaluation (10 random splits): mean ± std across seeds for each model and target.

| Target | Model | MAE | RMSE | $R^2$ | $\mathrm{Cov}_{90}$ | $\mathrm{Cov}_{95}$ |
|---|---|---|---|---|---|---|
| Y1 | Ridge | 2.097 ± 0.121 | 2.887 ± 0.161 | 0.917 ± 0.009 | n/a | n/a |
| Y1 | Random Forest | **0.339 ± 0.020** | **0.504 ± 0.024** | **0.997 ± 0.000** | n/a | n/a |
| Y1 | GPR | 0.536 ± 0.044 | 0.719 ± 0.063 | 0.995 ± 0.001 | 0.968 ± 0.014 | 0.984 ± 0.010 |
| Y1 | GPR (Numeric) | 0.450 ± 0.024 | 0.630 ± 0.033 | 0.996 ± 0.001 | **0.951 ± 0.013** | **0.977 ± 0.016** |
| Y2 | Ridge | 2.351 ± 0.186 | 3.283 ± 0.243 | 0.880 ± 0.015 | n/a | n/a |
| Y2 | Random Forest | 1.158 ± 0.118 | 1.876 ± 0.137 | 0.961 ± 0.006 | n/a | n/a |
| Y2 | GPR | 1.084 ± 0.114 | 1.502 ± 0.233 | 0.974 ± 0.008 | 0.907 ± 0.034 | **0.940 ± 0.028** |
| Y2 | GPR (Numeric) | **0.858 ± 0.109** | **1.287 ± 0.143** | **0.981 ± 0.004** | **0.903 ± 0.031** | 0.935 ± 0.025 |

Source: Author's elaboration.

**Table A2.** Post-hoc split conformal calibration check for the Gaussian-process model (calibration split from the training set, seed 42). Gap denotes the difference between empirical and nominal coverage.

| Target | $q_{0.90}$ | $q_{0.95}$ | $\mathrm{Cov}_{90}$ | $\mathrm{Cov}_{95}$ | Gap90 | Gap95 |
|---|---|---|---|---|---|---|
| Y1 | 1.692 | 2.392 | 0.948 | 0.974 | +0.048 | +0.024 |
| Y2 | 2.747 | 3.577 | 0.896 | 0.955 | −0.004 | +0.005 |

Source: Author's elaboration.

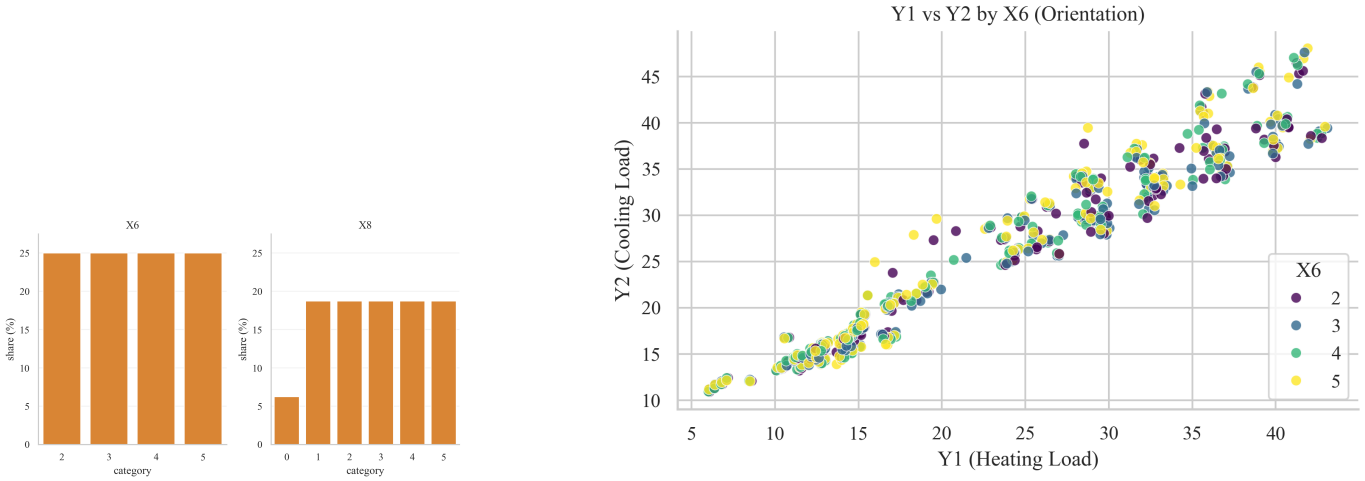



**Fig. A1.** Category counts for X6 and X8 (left) and the joint Y1/Y2 view colored by orientation X6 (right).

Source: Author's elaboration.

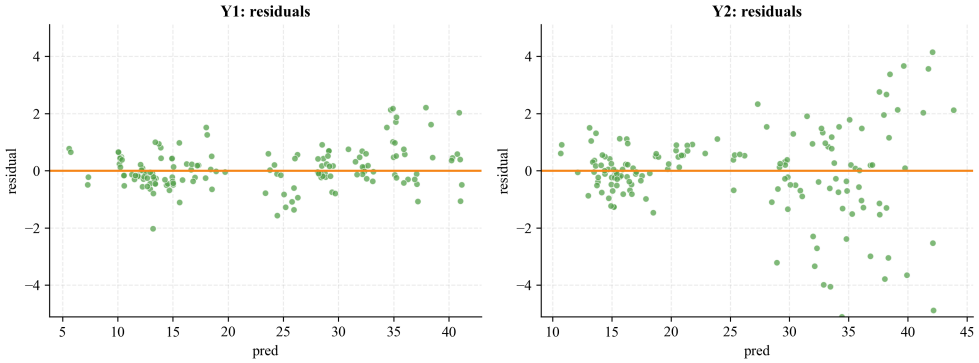



**Fig. A2.** Residuals versus predicted values for the Gaussian-process model on Y1 (Heating Load) and Y2 (Cooling Load).

Source: Author's elaboration.

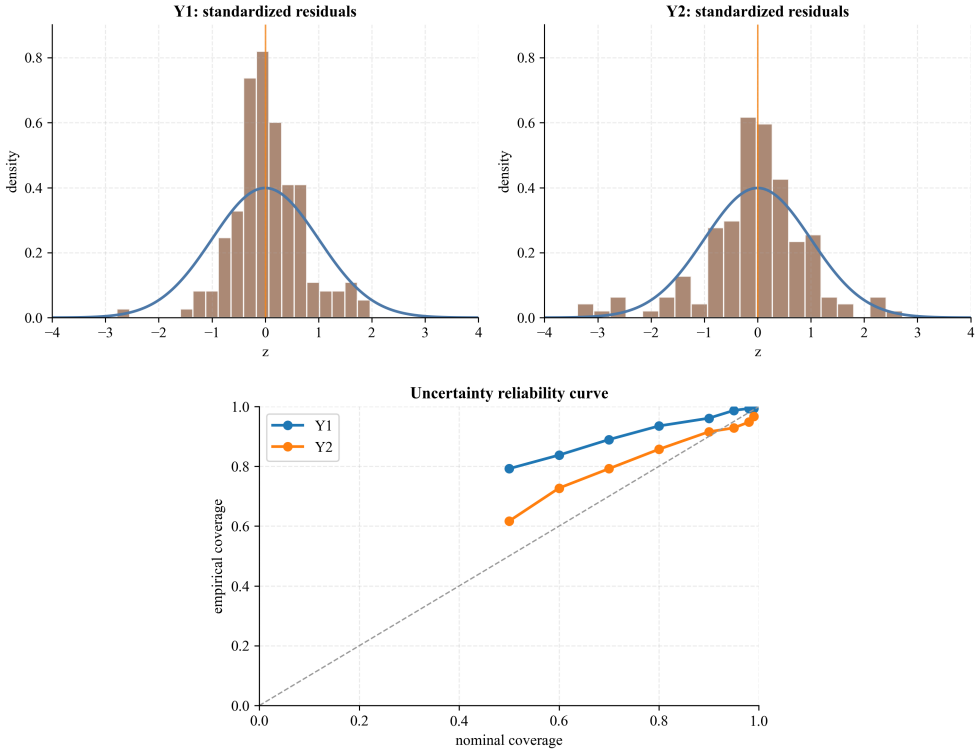



**Fig. A3.** Uncertainty diagnostics for the Gaussian-process model: standardized residuals (top) and reliability curve (bottom) comparing nominal versus empirical coverage.

Source: Author's elaboration.